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Surgical execution apparatus and surgical assistance device

Abstract

The present disclosure relates to the field of medical device technology, and specifically discloses a surgical execution apparatus and a surgical assistance device, the apparatus comprising a transmission cabin and a power cabin, the transmission cabin comprising a chassis assembly and first to third adapter transmission assemblies movable relative to the chassis assembly, the first to third adapter transmission assemblies comprising first to third transmission modules detachably connected to first to third adapter assemblies respectively; the power cabin being in transmission connection to the transmission cabin and configured to drive the first to third adapter assemblies via the first to third transmission modules. The apparatus enables more precise, stable and safe surgical execution operations through refined improvements in the transmission cabin.

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Background/Summary

(1) This application is a continuation of International Application No. PCT/CN2024/140574, filed on Dec. 19, 2024, which claims the priority to Chinese Patent Application No. CN202311773175.1 filed on Dec. 20, 2023, and entitled “SURGICAL EXECUTION APPARATUS AND SURGICAL ASSISTANCE DEVICE”, the entire disclosure of which is incorporated by reference herein.

TECHNICAL FIELD

(1) The present disclosure relates to the field of medical device technology, and in particular to a surgical execution apparatus and a surgical assistance device.

BACKGROUND

(2) Minimally invasive valve repair surgery is an advanced surgical approach for treating valvular heart diseases including those of mitral and tricuspid valves. Taking mitral valve repair surgery as

an example, the surgery is primarily performed via a transcatheter method, which can reduce surgical trauma and shorten recovery time. Specifically, the surgeon delivers the catheter to the mitral valve site through the femoral vein or via an apical puncture, where specialized instruments are used to repair the damaged valve. This surgical approach offers several advantages, including minimal trauma, reduced bleeding, and faster recovery, as well as effectively lowered incidence of postoperative complications.

(3) Traditional minimally invasive mitral valve repair surgery requires surgeons to manually operate invasive mitral valve repair instruments with their both hands, in order to for example hold handles, manipulate knobs, extend or retract the handles and push or pull levers, to complete the surgical procedures. In most cases, the surgical procedures need to be performed under the fluoroscopic guidance, thereby necessitating the surgeons to manually manipulate the instruments while wearing heavy lead protective gears for prolonged time, which significantly impacts the quality of surgeons' diagnosis and treatment activities, as well as their physical stamina and health, especially for experienced elderly or female surgeons. Certain complex surgical techniques or instruments, due to their high operational difficulty, demand a high level of technical skill and clinical experience from surgeons, and requires a relatively long learning curve for surgeons, which to some extent restricts the development of surgical techniques or the clinical application of instruments.

SUMMARY

(4) The present disclosure is directed to providing a surgical execution apparatus and a surgical assistance device for facilitating valve repair surgery for surgeons, so as to improve the operational accuracy and efficiency of valve repair instruments, reduce the physical demands on surgeons, and shorten the learning curves required for surgeons to master complex surgical techniques.

(5) To achieve this objective, the present disclosure provides the following technical solutions:

(6) A surgical execution apparatus includes a transmission cabin and a power cabin, the transmission cabin including a chassis assembly and a first adapter transmission assembly, a second adapter transmission assembly and a third adapter transmission assembly that are movable relative to the chassis assembly, and the power cabin being in transmission connection to the transmission cabin, where the first adapter transmission assembly is slidably attached to the chassis to slide along an adjustment direction that is parallel to a lengthwise direction of the transmission cabin, the second adapter transmission assembly is slidably attached to the first adapter transmission assembly to slide along the adjustment direction, and the third adapter transmission assembly is slidably attached to the second adapter transmission assembly to slide along the adjustment direction.

(7) As a preferred technical solution of the surgical execution apparatus, the power cabin includes a motor assembly including an assembly frame plate and a plurality of drive motors mounted on the assembly frame plate, with output ends of the drive motors extending through the assembly frame plate; and the transmission cabin includes an axial end fixing assembly provided at one end of the first adapter transmission assembly, the axial end fixing assembly comprising a fixing end plate and input end connection assemblies extending through and rotatably connected to the fixing end plate, where the input end connection assemblies and the drive motors are equal in number, and each input end connection assembly is coaxially and detachably connected to an output end of one of the drive motors.

(8) As a preferred technical solution of the surgical execution apparatus, the chassis assembly includes a bottom bracket and a guide rail connection plate slidably provided on the bottom bracket, and the first adapter transmission assembly includes a frame, with the axial end fixing assembly being fixed to one end of the frame, and the guide rail connection plate being fixed to the frame.

(9) As a preferred technical solution of the surgical execution apparatus, a first trapezoidal nut is installed on the bottom bracket, and a first trapezoidal screw is rotatably connected to the guide rail

connection plate to rotate around the adjustment direction, where the first trapezoidal nut is in transmission engagement with the first trapezoidal screw; and/or the second adapter transmission assembly includes a transmission support plate with a second trapezoidal nut being installed thereon, and a second trapezoidal screw is rotatably connected to the axial end fixing assembly to rotate around the adjustment direction, where the second trapezoidal nut is in transmission engagement with the second trapezoidal screw; and/or a third trapezoidal screw is rotatably connected to the transmission support plate to rotate around the adjustment direction, and the third adapter transmission assembly includes a slider connection plate with a third trapezoidal nut being installed thereon, where the third trapezoidal nut is in transmission engagement with the third trapezoidal screw.

(10) As a preferred technical solution of the surgical execution apparatus, the first trapezoidal screw is coaxially fixed to one of the input end connection assemblies via a first cross slider coupling and one of second cross slider couplings in sequence; and/or the second trapezoidal screw is coaxially fixed to one of the input end connection assemblies via one of the second cross slider couplings; and/or the third trapezoidal screw is coaxially fixed to one of the input end connection assemblies via one of telescopic universal couplings and one of the second cross slider couplings in sequence.

(11) As a preferred technical solution of the surgical execution apparatus, the first adapter transmission assembly includes a first transmission module including a plurality of first transmission structures, each including a first transmission shaft rotatable around the adjustment direction, with respective third cross slider couplings being coaxially fixed to both ends of the first transmission shaft, where one end of the first transmission shaft is coaxially fixed to one of the input end connection assemblies, and the other end is in transmission connection to one of first adapter output assemblies via a first bevel gear swivel.

(12) As a preferred technical solution of the surgical execution apparatus, the second adapter transmission assembly includes a second transmission module including a plurality of second transmission structures, each including a second transmission shaft rotatable around the adjustment direction, where one end of the second transmission shaft is coaxially fixed to one of the input end connection assemblies via one of telescopic universal couplings and one of second cross slider couplings in sequence, and the other end is in transmission connection to one of second adapter output assemblies via a second bevel gear swivel.

(13) As a preferred technical solution of the surgical execution apparatus, the third adapter transmission assembly includes a third transmission module including a plurality of third transmission structures, each including a ball spline assembly rotatable around the adjustment direction, where the ball spline assembly includes a ball spline outer shaft and a ball spline inner shaft in transmission engagement, the ball spline inner shaft being insertable into the ball spline outer shaft and coaxially fixed to one of the input end connection assemblies via a universal coupling, and the ball spline outer shaft being in transmission connection to one of third adapter output assemblies via a third bevel gear swivel.

(14) A surgical assistance device includes a robotic arm, a first adapter assembly, a second adapter assembly, a third adapter assembly and the above-mentioned surgical execution apparatus that is detachably connected to the robotic arm, where the first adapter transmission assembly includes a first transmission module detachably connected to the first adapter assembly, the second adapter transmission assembly includes a second transmission module detachably connected to the second adapter assembly, and the third adapter transmission assembly includes a third transmission module detachably connected to the second adapter assembly; the power cabin is configured to drive the first adapter assembly via the first transmission module, drive the second adapter assembly via the second transmission module, and drive the third adapter assembly via the third transmission module.

(15) As a preferred technical solution of the surgical assistance device, the transmission cabin

further includes an operation panel provided on the chassis assembly, the operation panel being communicatively connected to the first adapter assembly, the second adapter assembly and the third adapter assembly, and configured to display parameter information of the first adapter assembly, the second adapter assembly and the third adapter assembly.

Beneficial Effects

(16) The surgical execution apparatus individually drives the first, second, and third adapter assemblies, which can enhance the flexibility of their output motions, facilitating refined operations required for surgical techniques, thus ensuring more precise, stable and safe operations of the surgical execution apparatus. Through the split design of the transmission cabin for power transmission and the power cabin for power generation, the active components are separated from the passive components in the surgical execution apparatus, thereby not only eliminating the risk of cable breakage or wear during transmission and movement of the surgical execution apparatus and avoiding issues related to the loosening of active connectors due to prolonged use, but also improving the electromagnetic shielding performance by isolating the active components within the power cabin. The transmission structure of the transmission cabin is simple and compact, which can effectively reduce the structural size of the surgical execution apparatus, and can improve the transmission efficiency, thus contributing to more stable, accurate and reliable operations. The apparatus with above structural design also offers high scalability, allowing for the replacement of different adapters according to specific surgical technique requirements, thereby expanding the range of applicable surgical techniques. The above improvements in the surgical execution apparatus can lower the experience required for surgeons in operating surgical instruments, decrease their physical exertion, and mitigate the impact of radiation from the surgical execution apparatus on them, while facilitating the intelligent operation of minimally invasive surgical diagnosis and treatment. The surgical execution apparatus, when being used by surgeons for surgical techniques, can ensure the stability and accuracy of surgical operations, and improve efficiency thereof, thus shortening the procedure durations. The application of the surgical execution apparatus also contributes to improving the working environment for surgeons, reducing the physical demands on them, and shortening the learning curves required for them to master complex surgical techniques.

(17) The surgical assistance device can be connected to the valve repair instrument via adapters to drive the movements of the instrument, thereby assisting surgeons in operating the valve repair instrument and complete the valve repair surgery.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of the structure of the surgical execution apparatus according to an embodiment of the present disclosure;
- (2) FIG. 2 is a schematic diagram of the structure of the transmission cabin according to an embodiment of the present disclosure;
- (3) FIG. 3 is a cross-sectional view of the transmission cabin according to an embodiment of the present disclosure;
- (4) FIG. 4 is a schematic diagram of the structure of the chassis assembly according to an embodiment of the present disclosure;
- (5) FIG. 5 is a schematic diagram of the structure of the first adapter transmission assembly according to an embodiment of the present disclosure;
- (6) FIG. 6 is a schematic diagram of the structure of the axial end fixing assembly according to an embodiment of the present disclosure;
- (7) FIG. 7 is a cross-sectional view of the axial end fixing assembly according to an embodiment of

the present disclosure;

(8) FIG. 8 is a schematic diagram of the structure of the input end connection assembly according to an embodiment of the present disclosure;

(9) FIG. 9 is a cross-sectional view of the input end connection assembly according to an embodiment of the present disclosure;

(10) FIG. 10 is a schematic diagram of the structure of the first transmission unit for first adapter assembly according to an embodiment of the present disclosure, viewed from a first perspective;

(11) FIG. 11 is a schematic diagram of the structure of the first transmission unit for first adapter assembly according to an embodiment of the present disclosure, viewed from a second perspective;

(12) FIG. 12 is a schematic diagram of the structure of the second transmission unit for first adapter assembly according to an embodiment of the present disclosure;

(13) FIG. 13 is a schematic diagram of the structure of the long shaft output assembly for first adapter assembly according to an embodiment of the present disclosure;

(14) FIG. 14 is a schematic diagram of the structure of the short shaft output assembly for first adapter assembly according to an embodiment of the present disclosure;

(15) FIG. 15 is a schematic diagram of the structure of the axial end assembly for first adapter assembly according to an embodiment of the present disclosure;

(16) FIG. 16 is a schematic diagram of the structure of the second adapter transmission assembly according to an embodiment of the present disclosure, viewed from the first perspective;

(17) FIG. 17 is a schematic diagram of the structure of the second adapter transmission assembly according to an embodiment of the present disclosure, viewed from the second perspective;

(18) FIG. 18 is a schematic diagram of the cross-sectional structure of the second adapter transmission assembly according to an embodiment of the present disclosure;

(19) FIG. 19 is a schematic diagram of the structure of the second adapter output assembly according to an embodiment of the present disclosure;

(20) FIG. 20 is a schematic diagram of the cross-sectional structure of the second adapter output assembly according to an embodiment of the present disclosure;

(21) FIG. 21 is a schematic diagram of the structure of the dual bearing seat assembly according to an embodiment of the present disclosure;

(22) FIG. 22 is a cross-sectional view of the dual bearing seat assembly according to an embodiment of the present disclosure;

(23) FIG. 23 is a schematic diagram of the structure of the trapezoidal screw unit according to an embodiment of the present disclosure;

(24) FIG. 24 is a cross-sectional view of the structure of the trapezoidal screw unit according to an embodiment of the present disclosure;

(25) FIG. 25 is a schematic diagram of the structure of the third adapter transmission assembly according to an embodiment of the present disclosure;

(26) FIG. 26 is a schematic diagram of the structure of the third adapter transmission assembly according to an embodiment of the present disclosure excluding the third adapter mounting cover;

(27) FIG. 27 is a cross-sectional view of the third adapter transmission assembly according to an embodiment of the present disclosure;

(28) FIG. 28 is a cross-sectional view of the third adapter output assembly according to an embodiment of the present disclosure;

(29) FIG. 29 is a schematic diagram of the structure of the power cabin according to an embodiment of the present disclosure;

(30) FIG. 30 is a schematic diagram of the structure of the power cabin according to an embodiment of the present disclosure excluding the top cover;

(31) FIG. 31 is a cross-sectional view of the power cabin according to an embodiment of the present disclosure;

(32) FIG. 32 is a schematic diagram of the structure of the assembly frame plate according to an

embodiment of the present disclosure;

(33) FIG. 33 is a schematic diagram of the structure of the first transmission module and the drive motors according to an embodiment of the present disclosure;

(34) FIG. 34 is a schematic diagram of the structure of the valve repair system according to an embodiment of the present disclosure;

(35) FIG. 35 is a schematic diagram of the structure of the mitral valve repair instrument according to an embodiment of the present disclosure;

(36) FIG. 36 is a schematic diagram of the structure of the second transmission module and the drive motors according to an embodiment of the present disclosure;

(37) FIG. 37 is a schematic diagram of the structure of the third transmission module and the drive motors according to an embodiment of the present disclosure;

(38) FIG. 38 is a cross-sectional view of the catheter assembly in the first state according to an embodiment of the present disclosure;

(39) FIG. 39 is a cross-sectional view of the catheter assembly in the second state according to an embodiment of the present disclosure;

(40) FIG. 40 is a cross-sectional view of the catheter assembly in the third state according to an embodiment of the present disclosure;

(41) FIG. 41 is a cross-sectional view of a human body according to an embodiment of the present disclosure;

(42) FIG. 42 is a cross-sectional view of a human body, a guide wire and a sheath according to an embodiment of the present disclosure;

(43) FIG. 43 is a cross-sectional view of a human body and a catheter assembly according to an embodiment of the present disclosure;

(44) FIG. 44 is a cross-sectional view of a heart according to an embodiment of the present disclosure;

(45) FIG. 45 is a cross-sectional view of a heart, a guide wire and a sheath according to an embodiment of the present disclosure;

(46) FIG. 46 is a cross-sectional view of a heart and a catheter assembly in the first state according to an embodiment of the present disclosure;

(47) FIG. 47 is a cross-sectional view of a heart and a catheter assembly in the second state according to an embodiment of the present disclosure; and

(48) FIG. 48 is a cross-sectional view of a heart and a catheter assembly in the third state according to an embodiment of the present disclosure, where:

(49) **1000**: power cabin; **1100**: bottom shell; **1200**: top cover; **1300**: motor assembly; **1310**: drive motor; **1320**: assembly frame plate; **1400**: driver assembly; **2000**: transmission cabin; **2100**: chassis assembly; **2110**: bottom bracket; **2120**: chassis main beam; **2121**: first linear guide rail; **2122**: first nut fixing seat; **2123**: first trapezoidal nut; **2130**: guide rail connection plate; **2131**: first bearing seat; **2132**: first trapezoidal screw; **2140**: first cross slider coupling; **2180**: L-shaped support; **2190**: handle; **2191**: bedside operation panel; **2200**: first adapter transmission assembly; **2210**: frame; **2211**: second bearing seat; **2212**: second linear guide; **2220**: axial end assembly for first adapter assembly; **2221**: output end bearing seat; **2222**: output end shaft body; **2223**: sixth flanged bearing; **2224**: sixth elastic retaining ring; **2225**: output end shaft key; **2230**: first transmission unit for first adapter assembly; **2231**: first adapter assembly mounting plate; **2232**: first positioning pin; **2233**: first bevel gear swivel; **2240**: second transmission unit for first adapter assembly; **2241**: three-bearing support; **2242**: first transmission shaft; **2250**: third cross slider coupling; **2260**: axial end fixing assembly; **2261**: fixing end plate; **2262**: universal coupling; **2263**: ball spline inner shaft; **2264**: telescopic universal coupling; **2265**: second cross slider coupling; **2270**: input end connection assembly; **2271**: first flanged bearing; **2272**: first elastic retaining ring; **2273**: input end connector shaft; **2274**: first adapter assembly connection flange; **2275**: first spring; **2276**: first column head screw; **2280**: long shaft output assembly for first adapter assembly; **2281**: second

adapter connection flange; **2282**: second flanged bearing; **2283**: second elastic retaining ring; **2284**: long connector shaft; **2285**: second spring; **2286**: second column head screw; **2287**: long connector bearing seat; **2290**: short shaft output assembly for first adapter assembly; **2291**: third adapter connection flange; **2292**: third flanged bearing; **2293**: third elastic retaining ring; **2294**: short connector shaft; **2295**: third spring; **2296**: third column head screw; **2297**: short connector bearing seat; **2300**: second adapter transmission assembly; **2310**: transmission support plate; **2311**: slider fixing plate; **2312**: first slider; **2320**: second adapter mounting cover; **2321**: second positioning pin; **2330**: second adapter output assembly; **2331**: fourth adapter connection flange; **2332**: fourth flanged bearing; **2333**: fourth elastic retaining ring; **2334**: second adapter connector shaft; **2335**: fourth spring; **2336**: fourth column head screw; **2337**: middle adapter output bearing seat; **2341**: second nut fixing seat; **2342**: second trapezoidal nut; **2350**: trapezoidal screw unit; **2351**: third trapezoidal screw; **2352**: third bearing seat; **2360**: dual bearing seat assembly; **2361**: bearing seat body; **2362**: bearing cover; **2363**: fifth column head screw; **2364**: first rolling bearing; **2370**: second bevel gear swivel; **2380**: reinforcing rib; **2400**: third adapter transmission assembly; **2410**: slider connection plate; **2411**: second slider; **2412**: second rolling bearing; **2420**: third adapter mounting cover; **2421**: third positioning pin; **2430**: third adapter output assembly; **2431**: fifth adapter connection flange; **2432**: fifth flanged bearing; **2433**: fifth elastic retaining ring; **2434**: first three adapter connector shaft; **2435**: fifth spring; **2436**: fifth column screw; **2440**: ball spline assembly; **2441**: ball spline body; **2442**: ball spline outer shaft; **2450**: third bevel gear swivel; **2460**: third trapezoidal nut; **2470**: second trapezoidal screw; **2910**: first bellows cover; **2920**: second bellows cover; **2930**: third bellows cover; **8000**: mitral valve repair instrument; **8100**: guide wire; **8200**: sheath; **8300**: catheter assembly; **8310**: outer catheter; **8320**: middle catheter; **8330**: inner catheter; **8340**: clip; **8400**: first driving handle; **8500**: second driving handle; **8600**: third driving handle; **8700**: first adapter assembly; **8800**: second adapter assembly; **8900**: third adapter assembly; **9000**: human body; **9100**: right femoral vein; **9200**: left femoral vein; **9300**: inferior vena cava; **9400**: descending aorta; **9500**: superior vena cava; **9600**: right jugular vein; **9700**: left jugular vein; **9800**: subclavian vein; **9900**: heart; **9910**: right atrium; **9920**: atrial septum; **9930**: left atrium; **9940**: mitral valve; **9950**: left ventricle.

DETAILED DESCRIPTION

(50) The technical solution of the present disclosure will be described below clearly and completely with reference to the accompanying drawings. It should be understood the embodiments described herein are only a part of the possible embodiments of the present disclosure, and do not represent all possible embodiments. Based on the embodiments of the present disclosure, any other embodiments obtained by a person of ordinary skill in the art without any creative effort are also within the scope of protection of the present disclosure.

(51) In the description of the present disclosure, it should be noted that the terms such as “center”, “upper”, “lower”, “left”, “right”, “vertical”, “horizontal”, “inner”, “outer” indicate the relative positions or orientations based on the those as shown in the drawings, for the purpose of concise description of the present disclosure only, rather than indicating or implying that the device or element referred to must have a particular orientation, or be constructed and operated in a particular orientation. Therefore, these terms should not be interpreted as limitations to the present disclosure. Furthermore, the terms such as “first” and “second” are for a distinguishing purpose only, and should not be construed as indicating or implying relative importance. Accordingly, the terms “first position” and “second position” refer to two distinct positions. When a first feature is described as being “above”, “over” and “on” a second feature, it includes both the case where the first feature is directly above the second feature and the case where the first feature is obliquely above the second feature, or simply indicates that the first feature is at a higher level than the second feature. When the first feature is described as “below”, “under” and “beneath” the second feature, it includes both the case where the first feature is directly below the second feature and the case where the first feature is obliquely below the second feature, or simply indicates that the first feature is at a lower

level than the second feature.

(52) In the description of the present disclosure, it should be noted that, unless otherwise clearly specified and limited, the terms such as “install”, “connect” and “couple” should be comprehended in a broad sense, and may be comprehended as, for example, fixed or detachable connection, or integrated connection; a mechanical or electrical connection; a direct connection or an indirect connection through an intermediate medium; or a communication between interiors of two elements. The specific meanings of the above terms in the present disclosure may be understood by a person of ordinary skill in the art according to specific context.

(53) The embodiments of the present disclosure are described below in detail, and examples of the embodiments are shown in the accompanying drawings, in which the same or similar reference numerals throughout represent the same or similar elements or elements having the same or similar functions. The embodiments described below with reference to the accompanying drawings are exemplary and are for the purpose of explaining the present disclosure only, and should not be construed as limiting the present disclosure.

(54) As shown in FIGS. **1** to **40**, the present embodiments provide a surgical assistance device, including a robotic arm, a first adapter assembly **8700**, a second adapter assembly, a third adapter assembly and a surgical execution apparatus, where the surgical execution apparatus is connected to the robotic arm, the first adapter assembly **8700**, the second adapter assembly and the third adapter assembly. The surgical assistance device can be connected to the valve repair instrument through adapters to drive the movements of the instrument, thereby assisting surgeons in operating the valve repair instrument and complete the valve repair surgery.

(55) The present embodiments are described by taking the mitral valve repair instrument **8000** as an example. The mitral valve repair instrument **8000** includes an implant and a delivery device. The delivery device includes a catheter assembly **8300**, a first drive handle **8400**, a second drive handle **8500**, and a third drive handle **8600**. The catheter assembly **8300** includes an outer catheter **8310**, a middle catheter **8320** and an inner catheter **8330**, which are nested together from the outside in. The inner catheter **8330** has two ends, i.e., a working end and an output end, where the working end is equipped with an implant, which in the present embodiments is a clip **8340** that can switch between an unfolded state and a folded state. The clip **8340** in the folded state can be stored in the middle catheter **8320**. The clip **8340** has a clamping portion, which in the unfolded state of the clip **8340** forms a clamping space for holding the valve leaflet.

(56) The surgical execution apparatus includes a transmission cabin **2000** and a power cabin **1000**. The transmission cabin **2000** including a chassis assembly **2100** and a first adapter transmission assembly **2200**, a second adapter transmission assembly **2300** and a third adapter transmission assembly **2400** that are movable relative to the chassis assembly **2100**, the first adapter transmission assembly **2200** having a first transmission module detachably connected to the first adapter assembly **8700**, the second adapter transmission assembly **2300** having a second transmission module detachably connected to the second adapter assembly **8800**, and the third adapter transmission assembly **2400** having a third transmission module detachably connected to the third drive module. The power cabin **1000** is in transmission connection to the transmission cabin **2000**, and configured to drive the first adapter assembly **8700** via the first transmission module, drive the second adapter assembly **8800** via the second transmission module, and drive the third adapter assembly **8900** via the third transmission module.

(57) The surgical execution apparatus individually drives the first adapter assembly **8700**, the second adapter assembly **8800**, and the third adapter assembly **8900**, which can enhance the flexibility of their output motions, facilitating refined operations required for surgical techniques, thus ensuring more precise, stable and safe operations of the surgical execution apparatus. Through the split design of the transmission cabin **2000** for power transmission and the power cabin **1000** for power generation, the active components are separated from the passive components in the surgical execution apparatus, thereby not only eliminating the risk of cable breakage or wear during

transmission and movement of the surgical execution apparatus and avoiding issues related to the loosening of active connectors due to prolonged use, but also improving the electromagnetic shielding performance by isolating the active components within the power cabin **1000**. The transmission structure of the transmission cabin **2000** is simple and compact, which can effectively reduce the structural size of the surgical execution apparatus, and can improve the transmission efficiency, thus contributing to more stable, accurate and reliable operations. The apparatus with above structural design also offers high scalability, allowing for the replacement of different adapters according to specific surgical technique requirements, thereby expanding the range of applicable surgical techniques. The above improvements in the surgical execution apparatus can lower the experience required for surgeons in operating surgical instruments, decrease their physical exertion, and mitigate the impact of radiation from the surgical execution apparatus on them, while facilitating the intelligent operation of minimally invasive surgical diagnosis and treatment. The surgical execution apparatus, when being used by surgeons for surgical techniques, can ensure the stability and accuracy of surgical operations, and improve efficiency thereof, thus shortening the procedure durations. The application of the surgical execution apparatus also contributes to improving the working environment for surgeons, reducing the physical demands on them, and shortening the learning curves required for them to master complex surgical techniques.

(58) In the present embodiments, the first adapter transmission assembly **2200** is slidably attached to the chassis assembly **2100** to slide along an adjustment direction that is parallel to a lengthwise direction of the transmission cabin **2000**, the second adapter transmission assembly **2300** is slidably attached to the first adapter transmission assembly **2200** to slide along the adjustment direction, and the third adapter transmission assembly **2400** is slidably attached to the second adapter transmission assembly **2300** to slide along the adjustment direction.

(59) The above structural layout enables the first adapter transmission assembly **2200**, the second adapter transmission assembly **2300** and the third adapter transmission assembly **2400** movable relative to the chassis assembly **2100**. This layout is simple and reliable, achieving a rational arrangement of the transmission cabin **2000**, thereby reducing the space occupied by the transmission cabin **2000**, and decreasing the difficulty of adjusting positions of the first adapter transmission assembly **2200**, the second adapter transmission assembly **2300** and the third adapter transmission assembly **2400**, and minimizing the likelihood of positional conflicts among the components within the transmission cabin **2000**.

(60) Furthermore, the power cabin **1000** includes a motor assembly **1300**, which includes an assembly frame plate **1320** and a plurality of drive motors **1310** mounted on the assembly frame plate **1320**, where output ends of the drive motors **1310** extend through the assembly frame plate **1320**; the transmission cabin **2000** includes an axial end fixing assembly **2260** provided at one end of the first adapter transmission assembly **2200**, the axial end fixing assembly **2260** including a fixing end plate **2261** and input end connection assemblies **2270** extending through and rotatably connected to the fixing end plate **2261**; the input end connection assemblies **2270** and the drive motors **1310** are equal in number, and each input end connection assembly **2270** is coaxially and detachably connected to an output end of one drive motor **1310**.

(61) The design of matching the input end connection assemblies **2270** with the output ends of the drive motors **1310** enables smooth transmission between the power cabin **1000** and the transmission cabin **2000**, and achieves modular designs for both the power cabin **1000** and the transmission cabin **2000**, thereby enhancing the flexibility of the use of the surgical execution apparatus, thus facilitating the expansion of its range of applicable surgical techniques.

(62) In the present embodiments, the power cabin **1000** is installed on the right side of the transmission cabin **2000**. The chassis assembly **2100** is located at the bottom of the transmission cabin **2000**, the first adapter transmission assembly **2200** is installed above the chassis assembly **2100**, the second adapter transmission assembly **2300** is installed above the first adapter transmission assembly **2200**, the third adapter transmission assembly **2400** is installed on the right

side of the second adapter transmission assembly **2300**, and the axial end fixing assembly **2260** is provided on the right side of the transmission cabin **2000**. The axial end fixing assembly **2260** can be quickly coupled to the motor assembly **1300**, allowing for rapid attachment and detachment of the transmission cabin **2000** to and from the power cabin **1000**.

(63) The transmission cabin **2000** further includes a first bellows cover **2910**, a second bellows cover **2920**, and a third bellows cover **2930**, where the first bellows cover **2910** is fixedly connected to the first adapter transmission assembly **2200** on the left side and to the second adapter transmission assembly **2300** on the right side, the second bellows cover **2920** is fixedly connected to the second adapter transmission assembly **2300** on the left side and to the third adapter transmission assembly **2400** on the right side, and the third bellows cover **2930** is fixedly connected to the third adapter transmission assembly **2400** on the left side and to the first adapter transmission assembly **2200** on the right side. The first bellows cover **2910** is configured to provide sealing protection during relative linear motion between the first adapter transmission assembly **2200** and the second adapter transmission assembly **2300**; the second bellows cover **2920** is configured to provide sealing protection during relative linear motion between the second adapter transmission assembly **2300** and the third adapter transmission assembly **2400**; and the third bellows cover **2930** is configured to provide sealing protection during relative linear motion between the third adapter transmission assembly **2400** and the first adapter transmission assembly **2200**.

(64) Specifically, the power cabin **1000** further includes a bottom shell **1100** and a top cover **1200** buckled onto the top of the bottom shell **1100**. The bottom shell **1100** and the top cover **1200** enclose a storage space with a driver assembly **1400** and all the drive motors **1310** being installed therein, where the driver assembly **1400** is communicatively connected to the motor assembly **1300**.

(65) Furthermore, the chassis assembly **2100** includes a bottom bracket **2110** and a guide rail connection plate **2130** slidably mounted thereon, and the first adapter transmission assembly **2200** includes a frame **2210**, with the axial end fixing assembly **2260** being fixedly connected to one end of the frame **2210**, where the guide rail connection plate **2130** is fixedly connected to the frame **2210**. A first trapezoidal nut **2123** is installed on the bottom bracket **2110**, and a first trapezoidal screw **2132** is rotatably connected to the guide rail connection plate **2130** to rotate around an adjustment direction, where the first trapezoidal nut **2123** is in transmission engagement with the first trapezoidal screw **2132**; the second adapter transmission assembly **2300** includes a transmission support plate **2310** with a second trapezoidal nut **2342** being installed thereon, and a second trapezoidal screw **2470** is rotatably connected to the axial end fixing assembly **2260** to rotate around the adjustment direction, where the second trapezoidal nut **2342** is in transmission engagement with the second trapezoidal screw **2470**; and a third trapezoidal screw **2351** is rotatably connected to the transmission support plate **2310** to rotate around the adjustment direction, and the third adapter transmission assembly **2400** includes a slider connection plate **2410** with a third trapezoidal nut **2460** installed thereon, where the third trapezoidal nut **2460** is in transmission engagement with the third trapezoidal screw **2351**.

(66) The above structural design of the transmission cabin **2000** establishes the transmission relationship between the chassis assembly **2100**, the first adapter transmission assembly **2200**, the second adapter transmission assembly **2300** and the third adapter transmission assembly **2400**, so that the relative movements between the above components is enabled by driving the first trapezoidal screw **2132**, the second trapezoidal screw **2470** and the third trapezoidal screw **2351**, thus decreasing the difficulty of driving and ensuring stable, long-term operation of the transmission cabin **2000**.

(67) In the present embodiments, a chassis main beam **2120** is fixedly screwed to the bottom bracket **2110**, with the guide rail connection plate **2130** being slidably provided on the chassis main beam **2120**, where the first trapezoidal screw **2132** is fixed to the guide rail connection plate **2130**

at both ends via respective first bearing seats **2131** (each having a bearing), and the first trapezoidal nut **2123** is installed on the first nut fixing seat **2122** which is in turn fixedly connected to the chassis main beam **2120**. When the first trapezoidal nut **2123** rotates, since the first trapezoidal nut **2123** is installed on the chassis main beam **2120**, the entire first trapezoidal screw **2132** can be driven to move axially. Considering that the first trapezoidal screw **2132** is installed on the guide rail connection plate **2130**, the guide rail connection plate **2130** is in turn driven to slide axially. During the entire movement process, the first trapezoidal screw **2132** bears minimal radial load, with all radial loads being transferred to the chassis main beam **2120** via the slider engagement, allowing for reduced structural size of the chassis assembly **2100**, thus achieving a more compact transmission structure of the chassis assembly **2100**. Specifically, a pair of first linear guide rails **2121** extending along the adjustment direction is provided on the chassis main beam **2120**, where the guide rail connection plate **2130** is slidably provided on the first linear guide rails **2121**.

(68) Furthermore, the first trapezoidal screw **2132** is coaxially fixed to one of the input end connection assemblies **2270** via a first cross slider coupling **2140** and one of second cross slider couplings **2265** in sequence; the second trapezoidal screw **2470** is coaxially fixed to one of the input end connection assemblies **2270** via one of the second cross slider couplings **2265**; and the third trapezoidal screw **2351** is coaxially fixed to one of the input end connection assemblies **2270** via one of telescopic universal couplings **2264** and one of the second cross slider couplings **2265** in sequence.

(69) The above design establishes the transmission connection between the axial end fixing assembly **2260** and the first trapezoidal screw **2132**, the second trapezoidal screw **2470** and the third trapezoidal screw **2351**, which simplifies the transmission mode between the transmission cabin **2000** and the power cabin **1000**, and optimizes the specific structure of the transmission cabin **2000**, thereby facilitating the smooth operation of the surgical execution apparatus.

(70) For example, the other end of the frame **2210** is equipped with a first transmission unit for first adapter assembly **2230** including a first adapter assembly mounting plate **2231** fixed to the frame **2210**, where a plurality of first positioning pins **2232** for pinning and positioning the first adapter assembly **8700** are provided on the first adapter assembly mounting plate **2231** to protrude therefrom; a second adapter mounting cover **2320** is provided above the transmission support plate **2310**, where a plurality of second positioning pins **2321** for pinning and positioning the second adapter assembly **8800** are provided on the second adapter mounting cover **2320** to protrude from; and a third adapter mounting cover **2420** is provided above the slider connection plate **2410**, where a plurality of third positioning pins **2421** for pinning and positioning the third adapter assembly **8900** are provided on the third adapter mounting cover **2420** to protrude therefrom.

(71) Specifically, a second linear guide rail **2212** extending along the adjustment direction is provided on the top end of the frame **2210**, and a slider fixing plate **2311** is further fixed to the transmission support plate **2310**, with a first slider **2312** being fixed to the bottom of the slider fixing plate **2311**, where the first slider **2312** is slidably engaged with the second linear guide rail **2212**.

(72) In the present embodiments, the drive motor **1310** drives the second trapezoidal screw **2470** to rotate. Since the second trapezoidal nut **2342** is fixed in its circumferential direction, the second trapezoidal nut **2342** can drive the second adapter transmission assembly **2300** to translate. During this process, the telescopic universal coupling **2264** extends or retracts, thus enabling the horizontal movement under the constraint of the sliding connection. The drive motor **1310** drives the third trapezoidal screw **2351** to rotate. Since the third trapezoidal nut **2460** is fixed in its circumferential direction, the third trapezoidal nut **2460** can drive the third adapter transmission assembly **2400** to translate. During this process, the ball spline body **2441** and the ball spline inner shaft **2263** form a translational friction pair, thus enabling the horizontal movement under the constraint of the sliding connection. Specifically, the second trapezoidal nut **2342** is rotatably connected to the second nut fixing seat **2341**, and the second nut fixing seat **2341** is fixedly connected to the transmission

support plate **2310**.

(73) Specifically, the second trapezoidal screw **2470** passes through the second bearing seat **2211** and is rotatably coupled to it, where the second bearing seat **2211** is fixedly connected to the frame **2210**.

(74) In the present embodiments, the third trapezoidal screw **2351** belongs to the trapezoidal screw unit **2350** which further includes two third bearing seats **2352** fixed to the transmission support plate **2310**, with the third trapezoidal screw **2351** being rotatably connected to the two third bearing seats **2352** at both ends.

(75) The second adapter transmission assembly **2300** further includes a dual bearing seat assembly **2360** including a bearing seat body **2361** fixed to the transmission support plate **2310**, where a plurality of bearing caps **2362** are detachably connected to the bearing seat body **2361** via fifth column head screws **2363**, each bearing cap **2362** forming a bearing space together with the bearing seat body **2361**, where a first rolling bearing **2364** is installed in the bearing space, and the second rotating shaft is fixedly coupled to the inner ring of the first rolling bearing **2364**, so that the second rotating shaft is rotatably connected to the dual bearing seat assembly **2360**.

(76) Specifically, a reinforcing rib **2380** for improving structural strength is further connected between the slider fixing plate **2311** and the transmission support plate **2310**.

(77) In the prior art, the catheter adapter transmission assembly employs an electric cylinder for driving, with a synchronous belt being provided for transmission within the transmission chain. However, this design suffers from poor controllability, low transmission efficiency and challenges in manufacturability, thus it is difficult to meet the requirements for speed control and position control.

(78) In the present embodiments, the first adapter transmission assembly **2200** has a first transmission module including a plurality of first transmission structures, each including a first transmission shaft **2242** rotatable around the adjustment direction, with respective third cross slider couplings **2250** being coaxially fixed to both ends of the first transmission shaft **2242**, where one end of the first transmission shaft **2242** is coaxially fixed to one input end connection assembly **2270**, and the other end is in transmission connection to one first adapter output assembly via a first bevel gear swivel **2233**, where the first adapter assembly **8700** is in transmission-engagement with the output ends of all the first adapter output assemblies.

(79) The above structural design establishes transmission connection between the axial end fixing assembly **2260** and the first adapter output assembly, which enables the power cabin **1000** to drive the first transmission module, thus ensuring that the first adapter assembly **8700** can complete the predetermined motions.

(80) Specifically, there are three first transmission structures, and the first adapter output assembly includes one long shaft output assembly **2280** for first adapter assembly and two short shaft output assemblies **2290** for first adapter assembly. The first transmission module includes three first transmission shafts **2242** and two three-bearing supports **2241** arranged oppositely, and one end of each first transmission shaft **2242** is rotatably connected to one three-bearing support **2241** to rotate around its own axis. All the above-mentioned first transmission shafts **2242** and all the three-bearing supports **2241** form the second transmission unit **2240** for first adapter assembly.

(81) The torque of the drive motor **1310** is transmitted to one of the third cross slider couplings **2250** via the axial end fixing assembly **2260**, and then to the second transmission unit **2240** for first adapter assembly, and then to the first adapter assembly **8700** installed on the first transmission unit **2230** for first adapter assembly via the other third cross slider coupling **2250** and the first transmission unit **2230** for first adapter assembly to achieve torque transmission. The third cross slider couplings **2250** are used to compensate for the accumulated tolerance caused during the assembly of parts.

(82) Exemplarily, the second adapter transmission assembly **2300** has a second transmission module including a plurality of second transmission structures, each including a second

transmission shaft rotatable around the adjustment direction, where one end of the second transmission shaft is coaxially fixed to one input end connection assembly **2270** via one telescopic universal coupling **2264** and one second cross slider coupling **2265** in sequence, and the other end is in transmission connection to one second adapter output assembly **2330** via a second bevel gear swivel **2370**, where the second adapter assembly **8800** is in transmission-engagement with the output ends of all the second adapter output assemblies **2330**.

(83) The above structural design establishes transmission connection between the axial end fixing assembly **2260** and the second adapter output assemblies **2330**, which enables the power cabin **1000** to drive the second transmission module, thus ensuring that the second adapter assembly **8800** can complete the predetermined motions.

(84) The torque of the drive motor **1310** is transmitted to the second cross slider coupling **2265** via the axial end fixing assembly **2260**, and then to the telescopic universal coupling **2264**, and then to the second adapter output assembly **2330** via the second transmission shaft, thereby achieving torque transmission.

(85) In the present embodiments, the third adapter transmission assembly **2400** has a third transmission module including a plurality of third transmission structures, each including a ball spline assembly **2440** rotatable around the adjustment direction, where the ball spline assembly **2440** includes a ball spline outer shaft **2442** and a ball spline inner shaft **2263** in transmission engagement, the ball spline inner shaft **2263** being insertable into the ball spline outer shaft **2442** and coaxially fixed to one input end connection assembly **2270** via the universal coupling **2262**, the ball spline outer shaft **2442** being in transmission connection to a third adapter output assembly **2430** via the third bevel gear swivel **2450**, where the third adapter assembly **8900** is in transmission-engagement with the output ends of all the third adapter output assemblies **2430**.

(86) The above structural design establishes transmission connection between the axial end fixing assembly **2260** and the third adapter output assemblies **2430**, which enables the power cabin **1000** to drive the third transmission module, thereby ensuring that the third adapter assembly **8900** can complete the predetermined motions.

(87) The torque of the motor torque is transmitted to the universal coupling **2262** via the axial end fixing assembly **2260**, and then to the ball spline inner shaft **2263**, and the torque is transmitted to the ball spline outer shaft **2442** via the ball spline body **2441**, and then to the third adapter output assembly **2430** via the third bevel gear swivel **2450**, thereby achieving torque transmission.

(88) In summary, the first adapter transmission assembly **2200**, the second adapter transmission assembly **2300** and the third adapter transmission assembly **2400** are all driven by motors, thus enabling good controllability, and can output speed, position and even force as needed; in addition, the above structure is highly flexible, which enables replacement with different adapter transmission assemblies as needed for the transmission cabin **2000**, thereby meeting different surgical technique requirements.

(89) In the present embodiments, thirteen input-end connection assemblies **2270** are connected to the shaft-end fixing assembly **2260**, which are in turn connected to five universal couplings **2262**, three telescopic universal couplings **2264** and five third cross-slider couplings **2250** respectively.

(90) Further, the input-end connection assembly **2270** includes an input-end connector shaft **2273**, two first flanged bearings **2271**, a first elastic retaining ring **2272**, a first adapter assembly connection flange **2274**, a first spring **2275** and two first column head screws **2276**. Each first flanged bearing **2271** has an inner ring fitted with the input-end connector shaft **2273**, and an outer ring fitted with the fixing end plate **2261**. The first flanged bearing **2271** on the right side is axially positioned via the shoulder on the input-end connector shaft **2273**. The shoulder on **273** realizes axial positioning, and the first flanged bearing **2271** on the left side is axially positioned via the first elastic retaining ring **2272**. The first spring **2275** is located in the hole on the right side of the input connector shaft **2273**. The first adapter assembly connection flange **2274** is installed on the right side of the first spring **2275**, and is secured by the first column head screw **2276**. When

axially pressed, the first adapter assembly connection flange **2274** can be axially displaced under the elastic force of the first spring **2275**. In this way, transmission engagement can be achieved without adding additional couplings, thereby facilitating the assembly and disassembly of the input end connection assembly **2270**. Specifically, the first flanged bearing **2271** is a flanged deep groove ball bearing, and the first column head screw **2276** is an internal hexagon column head screw.

(91) Exemplarily, the long shaft output assembly **2280** for first adapter assembly includes a second adapter connection flange **2281**, two second flanged bearings **2282**, a second elastic retaining ring **2283**, a long connector shaft **2284**, a second spring **2285**, two second column screws **2286** and a long connector bearing seat **2287**. The second flanged bearing **2282** has an outer ring fitted with the long connector bearing seat **2287** that is fixed relative to the frame **2210**.

(92) In the present embodiments, the short shaft output assembly **2290** for first adapter assembly includes a third adapter connection flange **2291**, a third flanged bearing **2292**, a third elastic retaining ring **2293**, a short connector shaft **2294**, a third spring **2295**, a third column screw **2296** and a short connector bearing seat **2297**. The third flanged bearing **2292** has an outer ring fitted with the short connector bearing seat **2297** that is fixed relative to the frame **2210**.

(93) In the present embodiments, the third adapter output assembly **2430** includes a fifth adapter connection flange **2431**, two fifth flanged bearings **2432**, a fifth elastic retaining ring **2433**, a third adapter connector shaft **2434**, a fifth spring **2435** and two fifth column head screws **2436**. The fifth flanged bearing **2432** has an outer ring fitted with the third adapter mounting cover **2420** that is fixed relative to the slider connection plate **2410**. Specifically, a second slider **2411** is provided at the bottom of the slider connection plate **2410**, and is slidably engaged with the transmission support plate **2310**; one end of the ball spline outer shaft **2442** away from the ball spline inner shaft **2263** is rotatably connected to the slider connection plate **2410** via the second rolling bearing **2412**.

(94) In the present embodiments, the second adapter output assembly **2330** includes a fourth adapter connection flange **2331**, two fourth flanged bearings **2332**, a fourth elastic retaining ring **2333**, a second adapter connector shaft **2334**, a fourth spring **2335**, two fourth column screws **2336** and a middle adapter output bearing seat **2337**. The fourth flanged bearing **2332** has an outer ring fitted with the middle adapter output bearing seat **2337**, and the middle adapter output bearing seat **2337** is fixed relative to the second adapter mounting cover **2320**.

(95) In the present embodiments, one end of the first transmission shaft **2242** away from the axial end fixing assembly **2260** is rotatably connected to the first adapter assembly mounting plate **2231** via the axial end assembly **2220** for first adapter assembly. The axial end assembly **2220** for first adapter assembly includes an output end bearing seat **2221**, an output end shaft body **2222**, two sixth flanged bearings **2223**, a sixth elastic retaining ring **2224** and an output end shaft key **2225**. The output end bearing seat **2221** has an outer wall inserted through the first adapter assembly mounting plate **2231**, and the sixth flanged bearing **2223** has an outer ring fitted with the output end bearing seat **2221**. The output end shaft key **2225** is fixedly connected to the output end shaft body **2222** which is keyed to the first transmission shaft **2242** via the output end shaft key **2225**.

(96) The specific structures of the above components are generally similar, all featuring a rotational connection between the rotating shaft and the bearing seat via the flanged bearing. Moreover, the rotating shaft has a spring interposed between the connection flange and the connector shaft, thereby enabling the connection flange and the connector shaft to move close to or away from each other in the lengthwise direction. Due to space limitations, detailed descriptions of the specific structures of these components are omitted here.

(97) Exemplarily, the transmission cabin **2000** further includes a bedside operation panel **2191** provided on the chassis assembly **2100**, where the bedside operation panel **2191** is communicatively connected to the first adapter assembly **8700**, the second adapter assembly **8800** and the third adapter assembly **8900** and configured to display parameter information of the first adapter assembly **8700**, the second adapter assembly **8800** and the third adapter assembly **8900**. Configuration of the bedside operation panel **2191** enables surgeons to interactively operate the

device at the operating bed, thereby facilitating more convenient and efficient surgical operations. (98) Specifically, the chassis assembly **2100** further includes a handle **2190** for surgeons to hold, where the handle **2190** is fixedly connected to the bottom bracket **2110**, with the bedside operation panel **2191** being located above the handle **2190**.

(99) The present embodiments also provide a surgical assistance device including a robotic arm, a first adapter assembly **8700**, a second adapter assembly **8800**, a third adapter assembly **8900** and the above-mentioned surgical execution apparatus, where the chassis assembly **2100** of the surgical execution apparatus is detachably connected to the robotic arm. Specifically, the chassis assembly **2100** further includes an L-shaped support **2180** fixedly connected to the right side of the bottom bracket **2110**, where an end flange is provided at the end of the robotic arm, and detachably connected to the L-shaped support **2180**.

(100) The surgical assistance device enables refined improvements on the operation of the mitral valve repair instrument **8000** through the cooperation of the surgical execution apparatus and the adapter assemblies. The use of surgical assistance device to assist the completion of the operation is intended to improve the quality and efficiency of the operation, mitigate or even avoid the harm of radiation to the health of surgeons, and enables simpler and more convenient procedure, thereby ultimately benefiting a large number of medical workers and patients.

(101) The transmission cabin **2000** includes a chassis assembly **2100** and a first adapter transmission assembly **2200**, a second adapter transmission assembly **2300** and a third adapter transmission assembly **2400** that are movable relative to the chassis assembly **2100**. The first adapter transmission assembly **2200** is equipped with a first adapter assembly **8700** for connecting to the outer catheter **8310**, the second adapter transmission assembly **2300** is equipped with a second adapter assembly **8800** for connecting to the middle catheter **8320**, and the third adapter transmission assembly **2400** is equipped with a third adapter assembly **8900** for connecting to the output end of the inner catheter **8330**. The power cabin **1000** is in transmission connection to the transmission cabin **2000**, and configured to drive the outer catheter **8310** through the first adapter assembly **8700**, drive the middle catheter **8320** through the second adapter assembly **8800**, and drive the inner catheter **8330** through the third adapter assembly **8900**.

(102) The surgical assistance device can assist in valve repair surgery by driving and controlling the mitral valve repair instrument **8000**. The design of separately driving the outer catheter **8310**, the middle catheter **8320** and the inner catheter **8330** can enhance the flexibility of motions of the mitral valve repair instrument **8000**, thus facilitates refined operations of the mitral valve repair instrument **8000**. When used by surgeons to complete valve repair surgery, the surgical assistance device further ensures the stability and accuracy of the procedure.

(103) The transmission cabin **2000** is configured to be connected to the mitral valve repair instrument **8000** through the first adapter assembly **8700**, the second adapter assembly **8800** and the third adapter assembly **8900**. On the one hand, the above structural improvement enables the mitral valve repair instrument **8000** to be effectively connected and fixed to the transmission cabin **2000** through the adapter assemblies, so that the surgeons can quickly, conveniently and stably install the mitral valve repair instrument **8000** on the transmission cabin **2000** during the operation preparation stage, and after the operation, the surgeons can quickly remove the mitral valve repair instrument **8000** from the transmission cabin **2000**. On the other hand, the adapters enable the power of each power output shaft inside the transmission cabin **2000** to be stably and reliably transferred to each operating handle, operating knob and operating pull rod on the mitral valve repair instrument **8000**.

(104) Specifically, the mitral valve repair instrument **8000** further includes a first driving handle **8400**, a second driving handle **8500**, a third driving handle **8600**, a guide wire **8100** and a sheath **8200** located at the end of the guide wire **8100**. The surgical assistance device further includes a first adapter assembly **8700**, a second adapter assembly **8800**, and a third adapter assembly **8900**. The first driving handle **8400** is mounted on the first adapter assembly **8700** and is in transmission

connection to the first adapter assembly **8700**, where the first adapter assembly **8700** is mounted on the first adapter transmission assembly **2200** and the first driving handle **8400** is configured to drive the outer catheter **8310**. The second driving handle **8500** is mounted on the second adapter assembly **8800** and is in transmission connection to the second adapter assembly **8800**, where the second adapter assembly **8800** is mounted on the second adapter transmission assembly **2300** and the second driving handle **8500** is configured to drive the middle catheter **8320**. The third driving handle **8600** is mounted on the third adapter assembly **8900** and is in transmission connection to the third adapter assembly **8900**, where the third adapter assembly **8900** is mounted on the third adapter transmission assembly **2400** and the third driving handle **8600** is configured to drive the inner catheter **8330**.

(105) As shown in FIGS. **41** to **48**, the human body **9000** has a right femoral vein **9100**, a left femoral vein **9200**, an inferior vena cava **9300**, a descending aorta **9400**, a superior vena cava **9500**, a right jugular vein **9600**, a left jugular vein **9700**, a subclavian vein **9800**, and a heart **9900**. The heart **9900** has a right atrium **9910**, an atrial septum **9920**, a left atrium **9930**, a mitral valve **9940**, and a left ventricle **9950**.

(106) The present embodiment also provides a method for performing a valve repair surgery, which is applicable to the above-mentioned surgical assistance device, and includes: inserting the guide wire **8100** into the human body **9000**, so that the sheath **8200** enters through the inferior vena cava **9300** into the right atrium **9910** of the heart **9900**; puncturing the atrial septum **9920** using the sheath **8200**; removing the guide wire **8100** and the sheath **8200**, inserting the catheter assembly **8300** into the human body **9000**, so that the catheter assembly **8300** enters through the inferior vena cava **9300** into the right atrium **9910** of the heart **9900** and the outer catheter **8310** passes through the atrial septum **9920** and at least partially extends into the left atrium **9930**; controlling the outer catheter **8310**, the middle catheter **8320** and the inner catheter **8330** to perform their own motions, so that the clip **8340** extends out of the middle catheter **8320** and switches to the unfolded state; controlling the outer catheter **8310**, the middle catheter **8320** and the inner catheter **8330** to perform their own motions again, so that the clip **8340** penetrates the mitral valve **9940**, and the mitral valve **9940** is clamped by the clamping part until the clip **8340** captures the mitral valve **9940**. During the process of performing the valve repair surgery method, the catheter assembly **8300** is in the first state, the second state and the third state in sequence.

(107) The valve repair surgery execution method enables smooth completion of the repair surgery for the mitral valve **9940** by sequentially introducing the guide wire **8100** and the mitral valve repair instrument **8000** into the human body **9000**. The process is simple, reliable and has high operational stability, which facilitates simplification of the procedure of valve repair surgery and optimization of its steps. The above steps can ensure the stability and accuracy of the surgery, thereby improving surgical efficiency and shortening the procedural time. The above method enables further improved working environment of surgeons, reduced physical demands on surgeons, and shortened learning curves for surgeons to master complex surgical techniques.

(108) It is evident that the described embodiments of the present disclosure are merely illustrative examples provided to elucidate the present disclosure, and do not constitute limitations on the implementation of the present disclosure. A person of ordinary skill in the art can make various changes or modifications on the basis of the above description. It is neither necessary nor possible to exhaustively enumerate all possible implementations. Any modifications, equivalent substitutions, and improvements made within the spirit and principles of the disclosure are intended to be encompassed within the scope of the appended claims.

Claims

1. A surgical execution apparatus, comprising: a transmission cabin (**2000**) comprising a chassis assembly (**2100**) and further comprising a first adapter transmission assembly (**2200**), a second

adapter transmission assembly (2300) and a third adapter transmission assembly (2400) that are movable relative to the chassis assembly (2100); and a power cabin (1000) in transmission connection to the transmission cabin (2000), wherein the first adapter transmission assembly (2200) is slidably attached to the chassis assembly (2100) to slide along an adjustment direction that is parallel to a lengthwise direction of the transmission cabin (2000), the second adapter transmission assembly (2300) is slidably attached to the first adapter transmission assembly (2200) to slide along the adjustment direction, and the third adapter transmission assembly (2400) is slidably attached to the second adapter transmission assembly (2300) to slide along the adjustment direction.

2. The surgical execution apparatus according to claim 1, wherein the power cabin (1000) comprises a motor assembly (1300) comprising an assembly frame plate (1320) and a plurality of drive motors (1310) mounted on the assembly frame plate (1320), with output ends of the drive motors (1310) extending through the assembly frame plate (1320); and the transmission cabin (2000) further comprises an axial end fixing assembly (2260) provided at one end of the first adapter transmission assembly (2200), the axial end fixing assembly (2260) comprising a fixing end plate (2261) and input end connection assemblies (2270) extending through and rotatably connected to the fixing end plate (2261), wherein the input end connection assemblies (2270) and the drive motors (1310) are equal in number, and each of the input end connection assemblies (2270) is coaxially and detachably connected to an output end of one of the drive motors (1310).

3. The surgical execution apparatus according to claim 2, wherein the chassis assembly (2100) comprises a bottom bracket (2110) and a guide rail connection plate (2130) slidably provided on the bottom bracket (2110), and the first adapter transmission assembly (2200) comprises a frame (2210), with the axial end fixing assembly (2260) being fixed to one end of the frame (2210), and the guide rail connection plate (2130) being fixed to the frame (2210).

4. The surgical execution apparatus according to claim 3, wherein a first trapezoidal nut (2123) is installed on the bottom bracket (2110), and a first trapezoidal screw (2132) is rotatably connected to the guide rail connection plate (2130) to rotate around the adjustment direction, the first trapezoidal nut (2123) being in transmission engagement with the first trapezoidal screw (2132); and/or the second adapter transmission assembly (2300) comprises a transmission support plate (2310) with a second trapezoidal nut (2342) being installed thereon, and a second trapezoidal screw (2470) is rotatably connected to the axial end fixing assembly (2260) to rotate around the adjustment direction, the second trapezoidal nut (2342) being in transmission engagement with the second trapezoidal screw (2470); and/or a third trapezoidal screw (2351) is rotatably connected to the transmission support plate (2310) to rotate around the adjustment direction, and the third adapter transmission assembly (2400) comprises a slider connection plate (2410) with a third trapezoidal nut (2460) being installed thereon, the third trapezoidal nut (2460) being in transmission engagement with the third trapezoidal screw (2351).

5. The surgical execution apparatus according to claim 4, wherein the first trapezoidal screw (2132) is coaxially fixed to one of the input end connection assemblies (2270) via one first cross slider coupling (2140) and one of second cross slider couplings (2265) in sequence; and/or the second trapezoidal screw (2470) is coaxially fixed to one of the input end connection assemblies (2270) via one of the second cross slider couplings (2265); and/or the third trapezoidal screw (2351) is coaxially fixed to one of the input end connection assemblies (2270) via one of telescopic universal couplings (2264) and one of the second cross slider couplings (2265) in sequence.

6. The surgical execution apparatus according to claim 3, wherein the first adapter transmission assembly (2200) further comprises a first transmission module comprising a plurality of first transmission structures, each comprising a first transmission shaft (2242) rotatable around the adjustment direction, with respective third cross slider couplings (2250) being coaxially fixed to both ends of the first transmission shaft (2242), wherein one end of the first transmission shaft (2242) is coaxially fixed to one of the input end connection assemblies (2270) and the other end is

in transmission connection to one of first adapter output assemblies via a first bevel gear swivel (2233).

7. The surgical execution apparatus according to claim 3, wherein the second comprising a plurality of second transmission structures, each comprising a second transmission shaft rotatable around the adjustment direction, with one end of the second transmission shaft being coaxially fixed to one of the input end connection assemblies (2270) via one of telescopic universal couplings (2264) and one of second cross slider couplings (2265) in sequence, and the other end being in transmission connection to one of second adapter output assemblies (2330) via a second bevel gear swivel (2370).

8. The surgical execution apparatus according to claim 3, wherein the third adapter transmission assembly (2400) comprises a third transmission module comprising a plurality of third transmission structures, each comprising a ball spline assembly (2440) rotatable around the adjustment direction, wherein the ball spline assembly (2440) comprises a ball spline outer shaft (2442) and a ball spline inner shaft (2263) in transmission engagement, the ball spline inner shaft (2263) being insertable into the ball spline outer shaft (2442) and coaxially fixed to one of the input end connection assemblies (2270) via a universal coupling (2262), and the ball spline outer shaft (2442) being in transmission connection to one of third adapter output assemblies (2430) via a third bevel gear swivel (2450).

9. A surgical assistance device, comprising a robotic arm, a first adapter assembly, a second adapter assembly, a third adapter assembly and a surgical execution apparatus that is detachably connected to the robotic arm and comprises: a transmission cabin (2000) comprising a chassis assembly (2100) and further comprising a first adapter transmission assembly (2200), a second adapter transmission assembly (2300) and a third adapter transmission assembly (2400) that are movable relative to the chassis assembly (2100); and a power cabin (1000) in transmission connection to the transmission cabin (2000), wherein the first adapter transmission assembly (2200) is slidably attached to the chassis assembly (2100) to slide along an adjustment direction that is parallel to a lengthwise direction of the transmission cabin (2000), the second adapter transmission assembly (2300) is slidably attached to the first adapter transmission assembly (2200) to slide along the adjustment direction, and the third adapter transmission assembly (2400) is slidably attached to the second adapter transmission assembly (2300) to slide along the adjustment direction, wherein the first adapter transmission assembly (2200) comprises a first transmission module detachably connected to the first adapter assembly, the second detachably connected to the second adapter assembly, and the third adapter transmission assembly (2400) comprises a third transmission module detachably connected to the second adapter assembly; and the power cabin (1000) is configured to drive the first adapter assembly via the first transmission module, drive the second adapter assembly via the second transmission module, and drive the third adapter assembly via the third transmission module.

10. The surgical assistance device according to claim 9, wherein the transmission cabin (2000) further comprises an operation panel (2191) provided on the chassis assembly (2100), the operation panel (2191) being communicatively connected to the first adapter assembly, the second adapter assembly and the third adapter assembly, and configured to display parameter information of the first adapter assembly, the second adapter assembly and the third adapter assembly.

11. The surgical assistance device according to claim 9, wherein the power cabin (1000) comprises a motor assembly (1300) comprising an assembly frame plate (1320) and a plurality of drive motors (1310) mounted on the assembly frame plate (1320), with output ends of the drive motors (1310) extending through the assembly frame plate (1320); and the transmission cabin (2000) further comprises an axial end fixing assembly (2260) provided at one end of the first adapter transmission assembly (2200), the axial end fixing assembly (2260) comprising a fixing end plate (2261) and input end connection assemblies (2270) extending through and rotatably connected to the fixing end plate (2261), wherein the input end connection assemblies (2270) and the drive

motors (1310) are equal in number, and each of the input end connection assemblies (2270) is coaxially and detachably connected to an output end of one of the drive motors (1310).

12. The surgical assistance device according to claim 11, wherein the chassis assembly (2100) comprises a bottom bracket (2110) and a guide rail connection plate (2130) slidably provided on the bottom bracket (2110), and the first adapter transmission assembly (2200) comprises a frame (2210), with the axial end fixing assembly (2260) being fixed to one end of the frame (2210), and the guide rail connection plate (2130) being fixed to the frame (2210).

13. The surgical assistance device according to claim 12, wherein a first trapezoidal nut (2123) is installed on the bottom bracket (2110), and a first trapezoidal screw (2132) is rotatably connected to the guide rail connection plate (2130) to rotate around the adjustment direction, the first trapezoidal nut (2123) being in transmission engagement with the first trapezoidal screw (2132); and/or the second adapter transmission assembly (2300) comprises a transmission support plate (2310) with a second trapezoidal nut (2342) being installed thereon, and a second trapezoidal screw (2470) is rotatably connected to the axial end fixing assembly (2260) to rotate around the adjustment direction, the second trapezoidal nut (2342) being in transmission engagement with the second trapezoidal screw (2470); and/or a third trapezoidal screw (2351) is rotatably connected to the transmission support plate (2310) to rotate around the adjustment direction, and the third adapter transmission assembly (2400) comprises a slider connection plate (2410) with a third trapezoidal nut (2460) being installed thereon, the third trapezoidal nut (2460) being in transmission engagement with the third trapezoidal screw (2351).

14. The surgical assistance device according to claim 13, wherein the first trapezoidal screw (2132) is coaxially fixed to one of the input end connection assemblies (2270) via one first cross slider coupling (2140) and one of second cross slider couplings (2265) in sequence; and/or the second trapezoidal screw (2470) is coaxially fixed to one of the input end connection assemblies (2270) via one of the second cross slider couplings (2265); and/or the third trapezoidal screw (2351) is coaxially fixed to one of the input end connection assemblies (2270) via one of telescopic universal couplings (2264) and one of the second cross slider couplings (2265) in sequence.

15. The surgical assistance device according to claim 12, wherein the first adapter transmission assembly (2200) further comprises a first transmission module comprising a plurality of first transmission structures, each comprising a first transmission shaft (2242) rotatable around the adjustment direction, with respective third cross slider couplings (2250) being coaxially fixed to both ends of the first transmission shaft (2242), wherein one end of the first transmission shaft (2242) is coaxially fixed to one of the input end connection assemblies (2270) and the other end is in transmission connection to one of first adapter output assemblies via a first bevel gear swivel (2233).

16. The surgical assistance device according to claim 12, wherein the second comprising a plurality of second transmission structures, each comprising a second transmission shaft rotatable around the adjustment direction, with one end of the second transmission shaft being coaxially fixed to one of the input end connection assemblies (2270) via one of telescopic universal couplings (2264) and one of second cross slider couplings (2265) in sequence, and the other end being in transmission connection to one of second adapter output assemblies (2330) via a second bevel gear swivel (2370).

17. The surgical assistance device according to claim 12, wherein the third adapter transmission assembly (2400) comprises a third transmission module comprising a plurality of third transmission structures, each comprising a ball spline assembly (2440) rotatable around the adjustment direction, wherein the ball spline assembly (2440) comprises a ball spline outer shaft (2442) and a ball spline inner shaft (2263) in transmission engagement, the ball spline inner shaft (2263) being insertable into the ball spline outer shaft (2442) and coaxially fixed to one of the input end connection assemblies (2270) via a universal coupling (2262), and the ball spline outer shaft

(**2442**) being in transmission connection to one of third adapter output assemblies (**2430**) via a third bevel gear swivel (**2450**).
